



## PHYSICS

Max Marks: 62

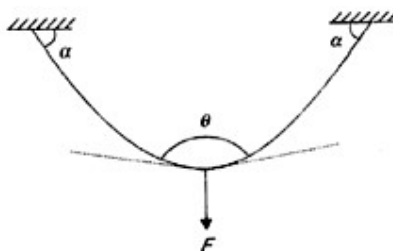
### SECTION – I

#### (SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

1. A rope of mass  $m$  is hung from a ceiling. The centre point is pulled down with a vertical force  $F$ . In equilibrium the tangent to the rope at its ends makes an angle  $\alpha$  with horizontal ceiling. The two tangents at the lower point make an angle of  $\theta$  with each other. Then  $\theta$  is

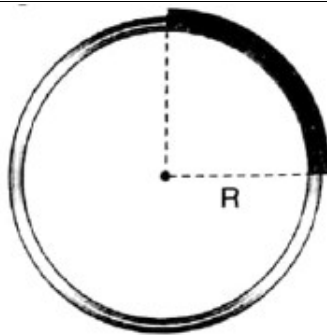


- A)  $\theta = 2 \tan^{-1} \left[ \left( 1 - \frac{mg}{F} \right) \cot \alpha \right]$

C)  $\theta = 2 \tan^{-1} \left[ \left( 1 + \frac{mg}{F} \right) \cot \alpha \right]$

B)  $\theta = \cot^{-1} \left[ \left( 1 + \frac{mg}{F} \right) \tan \alpha \right]$

D)  $\theta = 2 \tan^{-1} \left[ \left( 1 + \frac{F}{mg} \right) \tan \alpha \right]$
2. A uniform rope of length  $\frac{\pi R}{2}$  has been placed on fixed cylinder of radius  $R$  as shown in the fig. One end of the rope is at the top of the cylinder. The coefficient of friction between the rope and the cylinder is just enough to prevent the rope from sliding. Mass of the rope is  $M$ . Then maximum tension in the rope is

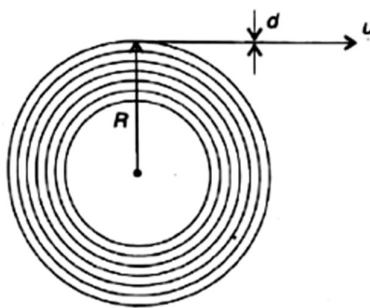


A)  $\left(\frac{\sqrt{2}-1}{\pi}\right)Mg$       B)  $2\left(\frac{\sqrt{2}+1}{\pi}\right)Mg$       C)  $\left(\frac{2\sqrt{2}-1}{\pi}\right)Mg$       D)  $2\left(\frac{\sqrt{2}-1}{\pi}\right)Mg$

3. A mass  $m$  moving with speed  $u$  in  $x$  direction collides elastically with a stationary mass  $2m$ . After the collision, it was found that both masses have equal  $x$  components of velocity. What angle does the velocity of mass  $2m$  make with the  $x$  axis?

A)  $45^\circ$       B)  $\tan^{-1}(\sqrt{2})$       C)  $30^\circ$       D)  $\tan^{-1}(1/\sqrt{2})$

4. A disc is free to rotate about an axis passing through its centre and perpendicular to its plane. The moment of inertia of the disc about its rotation axis is  $I$ . A light ribbon is tightly wrapped over it in multiple layers. The end of the ribbon is pulled out at a constant speed of  $u$ . Let the radius of the ribboned disc be  $R$  at any time and thickness of the ribbon be  $d$  ( $d \ll R$ ). Find the force ( $F$ ) required to pull the ribbon as a function of radius  $R$ .



A)  $\frac{Idu^2}{2\pi R^2}$       B)  $\frac{Idu^2}{2\pi R^4}$       C)  $\frac{Iu^2}{2\pi R^3}$       D)  $\frac{Idu^2}{\pi R^4}$

5. Due to rotation of the earth the direction of vertical at a place is not along the radius of the earth and actually makes a small angle  $\phi$  with the true vertical (i.e with radius). At what latitude ( $\theta$ ) is this angle  $\phi$  maximum? ( $\omega^2 R \ll g$ )

A)  $45^\circ$       B)  $0^\circ$       C)  $22.5^\circ$       D)  $67.5^\circ$

6. A cylindrical container having radius  $r$  has perforated wall. There are large number of uniformly spread small holes on the vertical wall occupying a fraction  $\eta = 0.02$  of the entire area of the wall. To maintain the water level at height  $H$  in the container, water is being fed to it at a constant rate  $Q$  ( $\text{m}^3/\text{s}$ ). Then  $Q$  is



- A)  $\frac{\eta}{3}\pi r^2\sqrt{2gH}$       B)  $\frac{4}{3}\pi\eta r\sqrt{2g}H^{3/2}$       C)  $\frac{2r\sqrt{2g}H^{3/2}}{3\eta\pi}$       D)  $\frac{2\pi r\sqrt{2g}H^{3/2}}{3\eta}$

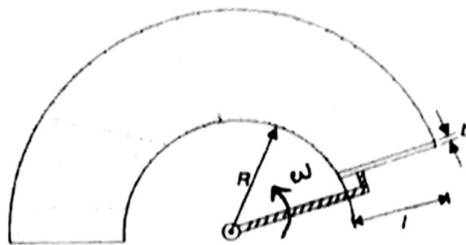
### SECTION – II

#### (MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases**

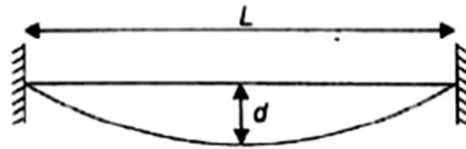
7. A car windshield wiper blade sweeps the wet windshield rotating at a constant angular speed of  $\omega$ .  $R$  is the radius of innermost arc swept by the blade. Length and width of the blade are  $l$  and  $b$  respectively. Coefficient of viscosity of water is  $\eta$ . Calculate the torque delivered by the motor to rotate the blade assuming that there is a uniform layer of thickness  $t$  on the glass surface.



- A)  $\frac{\eta b \omega R^3}{3t} \left( \left( 1 + \frac{l}{R} \right)^3 - 1 \right)$       B)  $\frac{\eta b \omega R^3}{2t} \left( \left( 1 + \frac{l}{R} \right)^2 - 1 \right)$
- C)  $\frac{\eta b \omega R^2 l}{3t}$       D)  $\frac{\eta b l \omega R^2}{2t}$

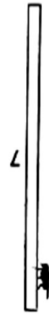
8. A wire having mass per unit length  $\mu$  and length  $L$  is fixed between two fixed vertical walls at a separation  $L$ . Due to its own weight the wire sags. The sag in the middle is

$d \ll L$ ). Assume that tension is practically constant along the wire, owing its small mass. Calculate the speed of the transverse wave on the wire.

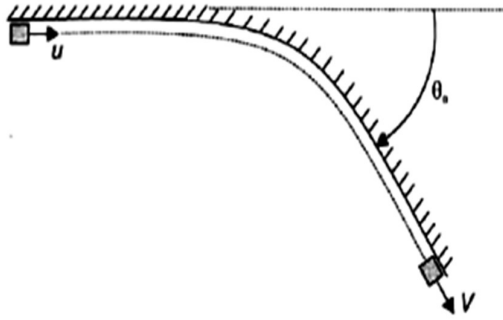


- A)  $L\sqrt{\frac{g}{4d}}$       B)  $L\sqrt{\frac{g}{8d}}$       C)  $L\sqrt{\frac{g}{2d}}$       D)  $L\sqrt{\frac{g}{16d}}$

9. A stick of length  $L$  is dropped from a high tower. An ant sitting at the lower end of the stick begins to crawl up at the instant the stick is released. Velocity of the ant relative to the stick remains constant and is equal to  $u$ . Assume that the stick remains vertical during its fall, and length of the stick is sufficiently long.

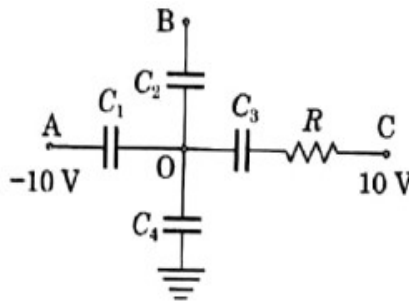


- A) Maximum height attained by the ant measured from its initial position is  $\frac{u^2}{2g}$
- B) Maximum height attained by the ant is greater than  $\frac{u^2}{2g}$
- C) After time  $\left(\frac{u}{g}\right)$  the ant is at the same height from where it has started
- D) After time  $\left(\frac{2u}{g}\right)$  the ant is at the same height from where it has started
10. A small object is sliding on a smooth horizontal floor along a vertical wall (figure shows the view from the top). The wall makes a smooth turn by an angle  $\theta_0$  (less than  $90^\circ$ ). Coefficient of friction between the wall and the block is  $\mu$ . Speed of the object before the turn is  $u$  and the final Velocity is  $V$ . [The turn is smooth without having any sharp corners.]



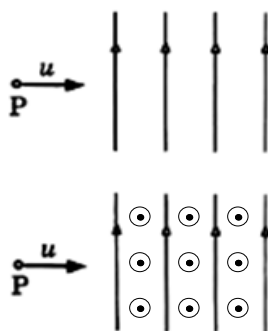
- A) Speed ( $V$ ) of object after completing the turn is  $ue^{-\mu\theta_0}$
- B) Speed ( $V$ ) of the object after completing the turn cannot be determined
- C) Speed ( $V$ ) does depend on the shape of the curve
- D) Speed ( $V$ ) does not depend on the shape of the curve

11. The figure shows part of a large network in which potentials of some of the points are shown. Each capacitor has a capacitor  $5\mu F$ . Which of the following statements is /are true?



- A) From the given information, potential of point O can be determined but that of B cannot be determined.
  - B) From the given information, potential of both the points O and B cannot be determined.
  - C) If charge on capacitor  $C_2$  were also specified, potential of point O can be determined but that of B cannot be determined
  - D) If charge on capacitor  $C_2$  were also specified, potentials of both the points O and B can be determined.
12. A positively charged particle P crosses a region of uniform electric field of intensity  $E$ . Velocity  $u$  of particle at the entry on the left side of the region is perpendicular to the electric field as shown in the first figure. Speed of the particle at exit on the right side is  $u_1$ . Now the experiment is repeated with a uniform magnetic field of induction  $B$  superimposed on the electric field as shown in the second figure, keeping entry

velocity of the particle unchanged. In this experiment, speed of the particle at exit on the right side is  $u_2$ . What can you conclude regarding the speeds  $u_1$  and  $u_2$ ?



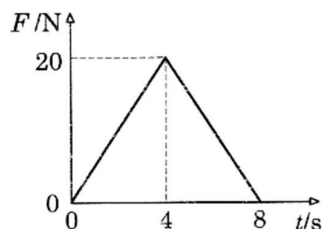
A) If  $u < \frac{E}{B}$  then  $u_2 < u_1$

B) If  $u = \frac{E}{B}$  then  $u_2 < u_1$

C) If  $u = \frac{E}{B}$  then  $u_2 = u_1$

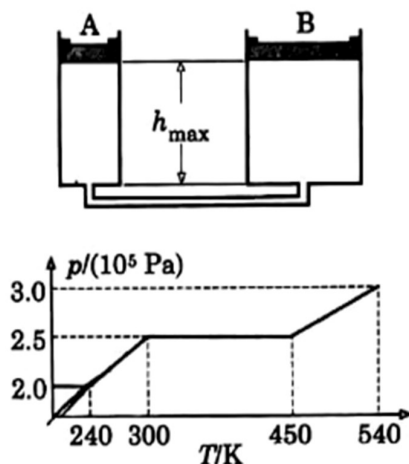
D) If  $u > \frac{E}{B}$  then  $u_2 < u_1$

13. A box of mass 4 kg placed on a horizontal floor, is acted upon by a horizontal force  $F$  that varies with time  $t$  as shown in the given graph. If coefficients of static and kinetic frictions are 0.25 each. On which of the following conclusions can you arrive?



- A) The box starts moving at the instant  $t = 2.0 \text{ s}$
- B) The maximum velocity acquired by the box is  $5.0 \text{ m/s}$
- C) The box stops at the instant  $t = 9.0 \text{ s}$
- D) Modulus of average power of the force  $F$  is more than that of the frictional force
14. Two fixed vertical cylinder vessels are connected at their bottoms by a thin tube. One mole of gaseous nitrogen is confined in the vessels under pistons A and B that can slide in the vessels without friction. Both the vessels have protrusions (stoppers) near their top on their inner surface to stop further upward motion of the pistons allowing a maximum height  $h_{\max} = 1.0 \text{ m}$  of columns of nitrogen gas in each of the vessels as shown in the figure. Outside the vessels is air at atmospheric pressure  $P_o = 10^5 \text{ pa}$ . In an experimental study pressure of the gas is measured over a wide range of temperature

and a pressure (p)- temperature (T)graph is prepared. A portion of the graph is shown in the adjacent figure. Mark the **CORRECT** option(s) (Take  $R = \frac{25}{3} \text{ J / mol / k}$ )



- A) Mass of piston A is 100 kg
- B) Mass of piston B is 75 kg
- C) Area of cross-section of piston A is  $0.01 \text{ m}^2$
- D) Area of cross-section of piston B is  $0.005 \text{ m}^2$

### SECTION – III

#### (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.**

#### Paragraph for Questions 15 and 16

A thin hollow spherical shell of radius  $R$  is filled with water. The temperature of atmosphere is  $T_s$  which is below  $0^\circ\text{C}$ . Firstly all water comes down to  $0^\circ\text{C}$  and then starts freezing from the outer surface towards the centre. Assuming that the thermal conductivity of shell material is very high. (Neglect the change in density due to freezing)

15. Find the time for the whole mass of the water to freeze. Thermal conductivity of ice is  $K$  and latent heat of fusion is  $L$ . ( $\rho$  = density)
- A)  $\frac{\rho L R^2}{6kT_s}$       B)  $\frac{2\rho L R^2}{kT_s}$       C)  $\frac{\rho L R^2}{3kT_s}$       D)  $\frac{\rho L R^2}{kT_s} \ln 2$
16. If total mass of water is  $M$  then
- A) Time taken for first  $7M/8$  mass to freeze is more than time taken for last  $M/8$  mass to freeze
- B) Time taken for first  $7M/8$  mass to freeze is same as time taken for last  $M/8$  mass to freeze

C) Time taken for first  $7M/8$  mass to freeze is less than time taken for last  $M/8$  mass to freeze

D) None of these

**Paragraph for Questions 17 and 18**

An amount of heat is slowly supplied to a bubble of radius  $r$  made of a soap solution of surface tension  $\sigma$ . If atmospheric pressure  $p_0$  outside the bubble remains constant, molar specific heat of the gas inside the bubble at constant pressure is  $C_p$  and universal gas constant is  $R$ , then

17. Find an expression for the molar specific heat of the heating process

A)  $C_p + R \left( \frac{3\sigma}{4p_0 r + 8\sigma} \right)$

B)  $C_p + R \left( \frac{3\sigma}{4p_0 r} \right)$

C)  $C_p + R \left( \frac{4\sigma}{3p_0 r + 8\sigma} \right)$

D)  $C_p + R \left( \frac{3\sigma}{p_0 r + 5\sigma} \right)$

18. Find an expression for the molar specific heat of the heating process if surrounding is vacuum

A)  $C_p + \frac{R}{2}$

B)  $C_p + R$

C)  $C_p + \frac{3R}{8}$

D)  $C_p + \frac{3R}{5}$

## SECTION – I

## (SINGLE CORRECT ANSWER TYPE)

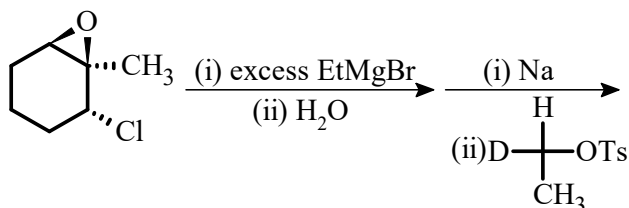
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

19. Which of the following is correctly arranged in increasing order of hydration energy?

- A)  $\text{ClO}_4^- < \text{ClO}_3^- < \text{ClO}^- < \text{ClO}_2^-$       B)  $\text{ClO}_4^- < \text{ClO}_3^- < \text{ClO}_2^- < \text{ClO}^-$   
 C)  $\text{ClO}_4^- < \text{ClO}^- < \text{ClO}_2^- < \text{ClO}_3^-$       D)  $\text{ClO}^- < \text{ClO}_2^- < \text{ClO}_3^- < \text{ClO}_4^-$

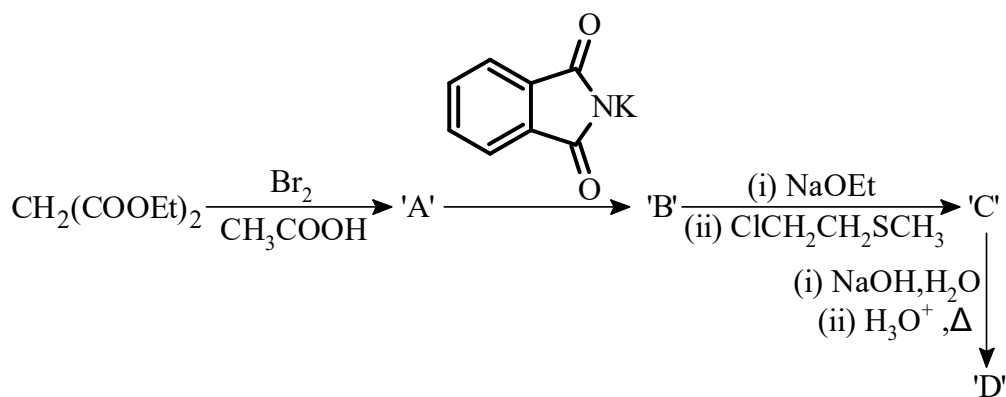
20.



Final organic product is

- A)
- B)
- C)
- D)

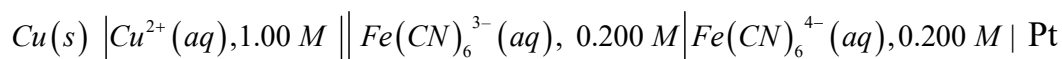
21.



Which of the following option is incorrect regarding Organic product 'D' is

- A) Essential amino acid.    B) Neutral amino acid.  
 C) Chiral compound      D) Insoluble in water.

22. The triple point of ammonia is 195.40 K at 0.006 bar, while its melting point is 195.42 K at 1 bar pressure. Which statement about ammonia is correct?
- A) Solid ammonia is denser than liquid ammonia at 195.4 K.  
 B) Ammonia vapor cannot coexist with liquid ammonia at equilibrium at 195.5 K.  
 C) Ammonia vapor cannot coexist with solid ammonia at equilibrium at 195.3 K.  
 D) Solid ammonia and liquid ammonia cannot coexist at equilibrium at 195.5 K.
23. The galvanic cell below has equal volumes in the two half-cells. As it is discharged at 25 °C, the cell potential gradually decreases to zero, whereupon the discharge stops. What is the concentration of  $\text{Cu}^{2+}(\text{aq})$  when the discharge stops?



| Half - Reaction                                                                                   | $E^0, V$ |
|---------------------------------------------------------------------------------------------------|----------|
| $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Cu}(s)$                                       | +0.337   |
| $\text{Fe}(\text{CN})_6^{3-}(\text{aq}) + e^- \rightarrow \text{Fe}(\text{CN})_6^{4-}(\text{aq})$ | +0.360   |

- A) 0.80M      B) 0.96M      C) 1.04M      D) 1.20 M
24. A spontaneous endothermic chemical reaction takes place at constant pressure and volume in a well-insulated container. As the reaction proceeds, what happens to the temperature of the system and to the internal energy of the reactants as they are transformed into products?
- A) Both the temperature and the reactant's internal energy increase.  
 B) The temperature increases while the reactant's internal energy decreases.  
 C) The temperature decreases while the reactant's internal energy increases.  
 D) Both the temperature and the reactant's internal energy decrease.

## SECTION – II

### (MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases**

25. Which of the following statements is/are correct.
- A) Inter molecular dehydration is a synthesis reaction in which larger molecules are formed

B) Hydrolysis reaction is a decomposition reaction forming different products from the reactant taken.

C) Hydration is a simple dipole-dipole attraction between solvent and solute molecules

D) Most of the hydrolysis reactions takes place due to polarisation of water molecules

26. Which of the following statements is/are correct for the complex having crystal field splitting energy of a complex is  $2.7 \times 10^{-19} J$  (plancks constant  $h = 6.62 \times 10^{-34} J.s$  and velocity of light  $C = 3 \times 10^8 m/s$ )

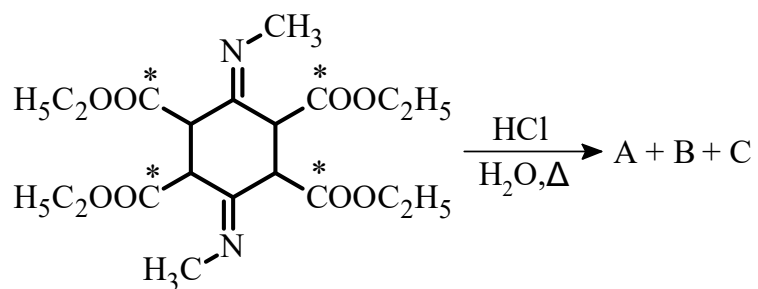
A) The wavelength of light (in nm) would be absorbed for the d – d transition is about 660 nm

B) The colour of the radiation absorbed by the complex corresponds to red

C) The solution of complex compound appear in green colour

D) The ligand can be considered as weak ligand

27.



A,B and C are organic compounds.

|                      | Iodoform test | Carbyl amine test |
|----------------------|---------------|-------------------|
| Organic compound 'A' | Negative      | Negative          |
| Organic compound 'B' | Negative      | Positive          |
| Organic compound 'C' | Positive      | Negative          |

Based on the given information, which of the following option is incorrect?

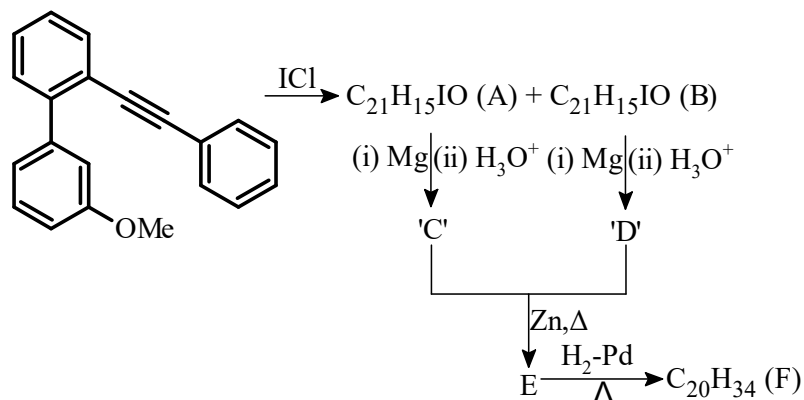
A) 'B' has labeled carbon.

B) 'A' do not show Tautomerism.

C) No of  $sp^3$  atoms :  $A > B > C$

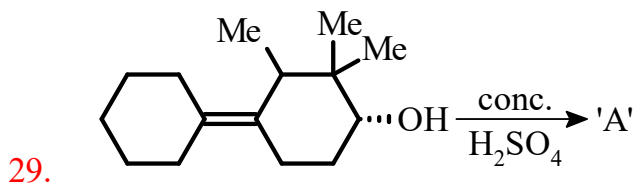
D) 'C' does not show conformational isomerism.

28.



Which of the following option is correct regarding 'F'?

- A) No of stereogenic centres in 'F' is 5.
- B) No of meso forms for 'F' is 0(zero).
- C) No of tertiary carbons in 'F' is 6.
- D) No of primary hydrogens in 'F' is 0(zero).



No of allylic hydrogens in 'A' is 10 and no of vinylic hydrogens in 'A' is 1.

Which of the following option is correct regarding 'A' is.

- A) No of geometrical isomers possible for 'A' is 2
  - B) No of stereo isomers possible for 'A' is 4
  - C) No of methyl groups in 'A' is 3
  - D) On reductive ozonolysis, 'A' gives two organic products.
30. A gas described by vander Waal's equation:
- A) Behave similar to an ideal gas in the limit of large molar volumes.
  - B) Behaves similar to an ideal gas is in limit of large pressures.
  - C) Is characterized by van de Waal's coefficients that are dependent on the identity of the gas but are independent of the temperature.

D) Has the pressure that is lower than the pressure exerted by the same gas behaving ideally

31. Choose the correct statements:
- A) Vacancy defects lower the density of the substance
  - B) Interstitial defects increase the density of the substance
  - C) Schottsky defects preserve the electrical neutrality of the crystal
  - D) Frenkel defects do not affect the density of the crystal.
32. The gaseous substance X was placed in a vacuumized vessel at  $25^{\circ}\text{C}$  and heated to  $500^{\circ}\text{C}$ . Over the course of the reaction, proceeding at a constant temperature, monitored by measuring the total pressure, when the degree of dissociation reached 51.7%, the gas in the vessel became equal in density with air ( $d_{\text{air}} = 29\text{g/cc}$  at the same conditions). The dependence of the reaction rate, as reaction being elementary, on time is given as  $\frac{1}{r} = 0.40(1.25t + 3.3)^3$  where  $r$  is expressed in bar / hours and  $t$  is the time
- When adding iodine vapor the reaction rate is increased 10,000 times, and the activation energy decreases by  $55\text{ kJ/mol}$ . Select the correct statement
- A) Molecular mass of X is greater than  $29\text{g/mol}$
  - B) Order of the reaction is 1.5
  - C) Pre-exponential factor in Arrhenius equation for catalytic reaction is more than non-catalytic reaction
  - D) Rate constant in the absence of the catalyst is  $2.5\text{ bar}^{-1}$

### SECTION – III

#### (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.**

#### Paragraph for Questions 33 and 34

Much of the world supply of platinum group metals is derived from the residues recovered from the electrolytic refining of copper and nickel. The residues when heated with aqua regia, the gold platinum and palladium go into solution. This is filtered off and to the filtrate on adding ferrous sulphate solution gold is precipitated. The platinum and palladium remain in solution.

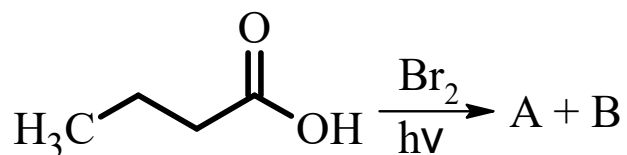
33. The role of ferrous sulphate in the precipitation of gold is
- A) Oxidant
  - B) Reductant
  - C) Complexing agent
  - D) Substituent

34. The structure of the compounds of platinum and palladium remained in the solution are
- A) Square planar and square planar
  - B) square planar and tetrahedral
  - C) Tetrahedral and square planar
  - D) Octahedral and square planar

**Paragraph for Questions 35 and 36**

In halogenations, electron-withdrawing groups greatly deactivate adjacent positions.

Compounds of the type  $Z-CH_2-CH_3$  are attacked predominantly or exclusively at the  $\beta$ -position when  $Z$  is  $COOH$ ,  $COCl$ ,  $COOR$ , or  $CX_3$ . Such compounds as acetic acid and acetyl chloride are not attacked at all. This is in sharp contrast to electrophilic halogenations, where only the  $\alpha$ -position is substituted. Halogen atoms are electrophilic radicals and look for positions of high electron density. Hydrogens on carbon atoms next to electron-withdrawing groups have low electron densities (because of the field effect of  $Z$ ) and are therefore shunned. Radicals that are not electrophilic do not display this behavior. For example, the methyl radical is essentially nonpolar and does not avoid positions next to electron-withdrawing groups.



‘A’ and ‘B’ are positional isomers. Based on the radical stability ‘A’ is major product and chiral.

35. 'A'  $\xrightarrow[\text{CaO}, \Delta]{\text{NaOH}}$  'X' Which of the following option is correct?

- A) 'X' cannot decolourise bromine water.
- B) 'X' cannot show structural isomerism.
- C) 'X' is soluble in aq  $\text{NaHCO}_3$ .
- D) 'X' does not show diastereomerism.

36. 'B'  $\xrightarrow[\text{CaO}, \Delta]{\text{NaOH}}$  'M'  $\xrightarrow[\text{Pt}]{\text{H}_2}$  'N' Which of the following option is incorrect?

- A) 'M' can decolourise bromine water.
- B) 'N' cannot decolourise Baeyer's reagent.
- C) 'M' is more acidic than 'N'.
- D) 'M' is more stable than 'N'

## SECTION – I

## (SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

37. Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ ,  $a + d = 1$ ,  $ad - bc = 1$  and  $I$  is the unit matrix then  $A^3 =$
- A)  $-I$                       B)  $I$                       C)  $2I$                       D)  $-2I$
38. If  $2a + 2b + 3c = \frac{1}{5}$  and  $a, b, c > 0$ , then maximum value of term independent of  $x$  in the expansion of  $(abx^{1/2} + cx^{-1/3})^{25}$  is
- A)  ${}^{25}C_{10}$                       B)  ${}^{25}C_{15}(35)^{25}$                       C)  ${}^{25}C_{15}\left(\frac{1}{35}\right)^{35}$                       D)  ${}^{25}C_{10}\left(\frac{1}{25}\right)^{35}$
39. Let  $\langle a_n \rangle$  be an arithmetic progression with common difference  $d (d \neq 0)$  and  $\langle b_n \rangle$  be a geometric progression with common ratio  $q$ , where  $q$  is a rational number and  $0 < q < 1$ . If  $a_1 = d$ ,  $b_1 = d^2$  and  $\frac{a_1^2 + a_2^2 + a_3^2}{b_1 + b_2 + b_3}$  is a positive integer then  $6q$  is
- A) 1                      B)  $1/2$                       C) 2                      D) 3
40.  $\int_{-2018\pi}^{2018\pi} \frac{(\cos x)^{2019} + 1}{(201720182019)^x + 1} dx$
- A) 0                      B)  $2018\pi$                       C)  $1009\pi$                       D)  $4036\pi$
41. If lines  $\frac{x-1}{2} = \frac{y-2}{x_1} = \frac{z-3}{x_2}$  and  $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$  lies in the same plane, then sum of the roots of  $x_1t^2 + (x_2 + 2)t + a = 0$  is
- A) -1                      B) -2                      C) 1                      D) 2
42. Area of the region bounded by  $x = 0$ ,  $x = 2$ ,  $y^2 = 4x$  and  $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$  when  $\alpha = 0$  and  $\alpha = 1$  is  $A_1$  and  $A_2$  respectively. Then  $|A_1 - A_2|$
- A) 1                      B) 2                      C) 4                      D) 9

## SECTION – II

## (MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases**

43. The range of the angle which a focal chord of a parabola  $y^2 = 4ax$  subtends at the vertex is

- A)  $\left(-\frac{\pi}{2}, -\tan^{-1}\left(\frac{4}{3}\right)\right)$  B)  $\left[\tan^{-1}\left(\frac{4}{3}\right), \frac{\pi}{2}\right)$   
C)  $\left(\frac{\pi}{2}, \pi - \tan^{-1}\left(\frac{4}{3}\right)\right]$  D)  $\left(\frac{\pi}{2}, \pi\right)$

44. Let  $\vec{a}$  and  $\vec{b}$  are two vectors  $|\vec{a}| = 2, |\vec{b}| = 3$  and  $\vec{a} \cdot \vec{b} = 0$ . If  $\vec{c}$  is a vector such that  $\vec{c} \times \vec{a} = \vec{b}$ .

The minimum value of  $2|\vec{c} - \vec{a}|$  is 'L' let  $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$  be non zero, non coplanar vectors.

Such that  $\vec{\gamma} \cdot \vec{\beta} = 0$  and  $\vec{\alpha} + (\vec{\gamma} \times \vec{\beta}) = \vec{\gamma} \times \vec{\alpha}$ . If angle between  $\vec{\alpha}$  &  $\vec{\beta}$  lies in  $\left[\frac{\pi}{6}, \frac{\pi}{4}\right]$ .

Maximum of  $|\vec{\gamma}|^2 = M$ . Minimum of  $|\vec{\gamma}|^2 = N$ . Then

- A)  $L = M$  B)  $L = M^2$   
C)  $L = \frac{M + N}{2}$  D)  $L = 3N$

45. The differential equation  $\{xy^3(1 + \cos x) - y\}dx + xdy = 0$  represents the curve

$$\frac{x^2}{ay^2} = \frac{x^3}{b} + x^2 \sin x + cx \cos x - d \sin x + k \text{ then}$$

- A)  $a = 3$  B)  $b = 3$   
C)  $a + b + c + d = 5$  D)  $c = 2$

46. If  $L = \lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^n \pi \sin \frac{k\pi}{2n}}$  (where  $\prod_{k=1}^{\infty} a_k = a_1 \cdot a_2 \dots$ ) then which of the following is/are correct

A) The number of all real valued continuously differentiable functions  $f$  on the real line such that for all  $x$ ,  $(f(x))^2 = \int_0^x ((f(t))^2 + (f'(t))^2) dt + 2018$  is  $4L$ .

B) The number of all real valued continuously differentiable functions  $f$  on the real line such that for all  $x$ ,  $(f(x))^2 = \int_0^x ((f(t))^2 + (f'(t))^2) dt + 2018$  is  $2L$ .

C) Number of elements in the range of  $g(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} \frac{1}{x} + \operatorname{cosec}^{-1} x + \sec^{-1} x + \tan^{-1} x$  is  $4L$

D) Number of elements in the range of  $g(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} \frac{1}{x} + \operatorname{cosec}^{-1} x + \sec^{-1} x + \tan^{-1} x$  is  $2L$

47. If  $\cot(3\alpha - 7\beta)\tan(5\beta - \alpha) = 1$ ,  $\alpha \in (0, \pi)$ ,  $\beta \in \left(0, \frac{\pi}{3}\right)$  and  $L = 2\sqrt{2} \sin\left(\frac{\alpha - 3\beta}{3}\right)$ , then

which is/are correct

A) Number of integral values of  $L$  is 2.

B) Number of integral values of  $L$  is 3

C) Number of irrational values of  $L$  is 4

D) Number of irrational values of  $L$  is 3

48. Which of the following is/are true

A) If  $z$  is a complex number and satisfying  $|z - 2 - i| = |z| \left| \sin\left(\frac{\pi}{4} - \arg(z)\right) \right|$  then locus of 'z' is a parabola.

B) If  $z$  is a complex number and satisfying  $|z - \cos^{-1}(\cos 12)| - |z - \sin^{-1}(\sin 12)| = 8(\pi - 3)$  then locus of 'z' is a hyperbola.

C) If  $z$  is a complex number and satisfying  $z^2 + az + a^2 = 0$  then  $|\arg(z)| = 2\pi/3 (a > 0)$

D) If  $z$  is a complex number and satisfying  $z^2 + az + a^2 = 0$  then  $\arg(z) = \frac{\pi}{3} (a \in R)$

49. Consider a system of linear equations  $a_i x + b_i y + c_i z = d_i$  (where  $a_i, b_i, c_i \neq 0$  and  $i = 1, 2, 3$ ) and  $(\alpha, \beta, \gamma)$  is its unique solution and  $a_1 = a_2 = a_3 = k, d_1 = d_2 = d_3 = k^2 (k \neq 0)$  and  $\alpha + \beta + \gamma = 2$  then

A)  $\lim_{x \rightarrow 0} \left( \frac{1}{\sin^2 x} - \frac{\ln(1+x^2)}{x^2 \sin^2 x} \right) = \frac{k}{4}$

B)  $k = 1$

C)  $k = 2$

D) Point  $(\alpha, \beta, \gamma)$  lie on z-axis.

50. Which of the following is/are true

A)  $f(x) = \begin{cases} \sin x \left| \sin \frac{\pi}{x} + \cos \frac{\pi}{x} \right| & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ , then  $f$  is not differentiable at  $x = 4/23$ .

B)  $g(x) = \begin{cases} x^2 + 4x & -3 \leq x \leq 0 \\ -\sin x & 0 < x \leq \frac{\pi}{2} \\ -\cos x - 1 & \frac{\pi}{2} < x \leq \pi \end{cases}$ , then  $x = \pi$  is the point of global maxima

C) Sum of all the solutions of the equation  $\sin \left( \pi \left( \log_3 \frac{1}{x} \right) \right) = 0$  in  $(0, 2\pi)$  is  $\frac{9}{2}$ .

D)  $\int x \cdot \left( \sum_{k=1}^{\infty} \frac{\pi \cos \frac{x}{2^k}}{2^k} \right) dx = \cos x + c$  (where  $C$  is integral constant) &

(where  $\sum_{k=1}^{\infty} a_k = a_1 + a_2 + \dots$ )

## SECTION – III

## (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.**

**Paragraph for Questions 51 and 52**

Football teams  $T_1$  and  $T_2$  have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of  $T_1$  winning, drawing and losing a game against  $T_2$  are  $\frac{1}{2}$ ,  $\frac{1}{6}$  and  $\frac{1}{3}$  respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams  $T_1$  and  $T_2$  respectively after two games.

51.  $P(X > Y)$  is

A)  $\frac{1}{4}$

B)  $\frac{5}{12}$

C)  $\frac{1}{2}$

D)  $\frac{7}{12}$

52.  $P(X = Y)$  is

A)  $\frac{11}{36}$

B)  $\frac{1}{3}$

C)  $\frac{13}{36}$

D)  $\frac{1}{2}$

**Paragraph for Questions 53 and 54**

If  $\overline{AB}$  and  $\overline{CD}$  are two perpendicular focal chords other than latus rectum of an ellipse

$$\frac{x^2}{4} + \frac{y^2}{3} = 1 \text{ with foci } S \text{ and } S'. \overline{AB} \text{ and } \overline{CD} \text{ both are passing through focus } S.$$

53. If  $AB = l_1$  and  $CD = l_2$  then  $12\left(\frac{1}{l_1} + \frac{1}{l_2}\right) =$

- A) 4                      B) 25                      C) 7                      D) 16

54. Value of semi perimeter of triangle  $S'AB$  is

- A) 4                      B) 8                      C) 7                      D) 16

**KEY SHEET**

|    |            |    |             |    |             |    |             |    |             |
|----|------------|----|-------------|----|-------------|----|-------------|----|-------------|
| 1  | <b>C</b>   | 2  | <b>D</b>    | 3  | <b>A</b>    | 4  | <b>B</b>    | 5  | <b>A</b>    |
| 6  | <b>B</b>   | 7  | <b>A</b>    | 8  | <b>B</b>    | 9  | <b>AD</b>   | 10 | <b>AD</b>   |
| 11 | <b>BD</b>  | 12 | <b>ABD</b>  | 13 | <b>ABCD</b> | 14 | <b>ABCD</b> | 15 | <b>A</b>    |
| 16 | <b>B</b>   | 17 | <b>C</b>    | 18 | <b>A</b>    |    |             |    |             |
| 19 | <b>B</b>   | 20 | <b>C</b>    | 21 | <b>D</b>    | 22 | <b>A</b>    | 23 | <b>C</b>    |
| 24 | <b>C</b>   | 25 | <b>ABCD</b> | 26 | <b>ABCD</b> | 27 | <b>ABD</b>  | 28 | <b>ABCD</b> |
| 29 | <b>ABC</b> | 30 | <b>AC</b>   | 31 | <b>ABCD</b> | 32 | <b>ABC</b>  | 33 | <b>B</b>    |
| 34 | <b>D</b>   | 35 | <b>D</b>    | 36 | <b>D</b>    |    |             |    |             |
| 37 | <b>A</b>   | 38 | <b>C</b>    | 39 | <b>D</b>    | 40 | <b>B</b>    | 41 | <b>B</b>    |
| 42 | <b>A</b>   | 43 | <b>C</b>    | 44 | <b>AD</b>   | 45 | <b>BD</b>   | 46 | <b>AC</b>   |
| 47 | <b>BC</b>  | 48 | <b>AC</b>   | 49 | <b>AC</b>   | 50 | <b>ABC</b>  | 51 | <b>B</b>    |
| 52 | <b>C</b>   | 53 | <b>C</b>    | 54 | <b>B</b>    |    |             |    |             |